
Pre-Registration Package:

Testable Predictions Derived from the Salience-Centered Paradigm (Levels I–III)

Signal Processing · Auditory Perception · Associative Memory Architecture

Author: Flyxion (independent researcher)
Components: Marine Salience Algorithm; HCF1 Compression; MEM|8
Scope: Six falsifiable predictions (Sets A, B, C)
Suitable for: OSF, AsPredicted, Registered Reports tracks
Date: May 2026

***Scope of this document.** This package concerns only the empirically tractable claims within Levels I–III of the Salience-Centered Paradigm (signal processing, compression architecture, associative memory). Level IV claims (consciousness, authenticity loss, and related constructs) are not pre-registered here; they require prior operationalization before experimental design is possible. The goal of this document is not to validate the broader paradigm but to derive risky, testable predictions from it — predictions whose failure would carry genuine evidential weight.*

Abstract

The Saliency-Centered Paradigm is a research programme proposing that temporal coherence — specifically, structured micro-temporal variation (jitter) in harmonic oscillators — constitutes signal-level information rather than noise, and that this insight motivates novel approaches to perceptual saliency estimation, audio compression, and phase-based associative memory. The paradigm spans four levels of increasing abstraction: temporal signal processing (Level I), compression architecture (Level II), memory architecture (Level III), and speculative claims about consciousness and authenticity (Level IV).

This pre-registration package concerns exclusively Levels I–III, which generate six falsifiable predictions amenable to standard experimental and computational methods. For Levels I and II, predictions address (A1) perceptual detection of temporal jitter in harmonic complexes under spectrally controlled conditions, (A2) incremental predictive power of a coherence metric in auditory saliency ratings, (B1) linear predictability of upper harmonic envelopes from the fundamental in acoustic instruments, and (B2) bitrate efficiency and perceptual transparency of the HCF1 compression scheme. For Level III, predictions address (C1) quadratic storage scaling in phase-encoded associative memory relative to a Hopfield baseline, and (C2) resistance to catastrophic forgetting under sequential learning.

Each template specifies a dependent variable, null hypothesis, statistical criterion, sample size with power justification, analysis pipeline, and confirmatory/exploratory distinction. The package follows open-science norms for pre-registration and is intended to be filed at OSF, AsPredicted, or an equivalent repository. Negative results are treated as substantive evidence; falsification of individual predictions is explicitly distinguished from falsification of the paradigm as a whole.

Keywords: auditory saliency; temporal coherence; harmonic compression; associative memory; pre-registration; falsifiability; signal detection theory; open science.

Contents

1	Notation and Terminology	4
2	Scientific Positioning	5
2.1	Prediction Set A — Psychoacoustics and Temporal Coding	5
2.2	Prediction Set B — Perceptual Audio Coding and Harmonic Modelling	5
2.3	Prediction Set C — Associative Memory and Attractor Networks	5
3	Cross-Level Architectural Unity	5
4	Explainability as an Independent Contribution	6
5	Operational Boundary Conditions	6
6	Statistical Philosophy and Confirmatory Logic	7
6.1	Confirmatory versus exploratory analyses	7
6.2	Effect thresholds and estimation	7
6.3	Signal detection theory (Predictions A1, A2)	7
6.4	Incremental regression (Prediction A2)	7
6.5	Replication priority	7
7	Relation to Existing Compression Paradigms	7
8	Computational Complexity and Scalability of MEM 8	8
8.1	Storage complexity	8
8.2	Retrieval complexity	8
8.3	Robustness under noise	9
8.4	Scalability	9
9	Preamble and Epistemic Framework	9
10	Prediction Set A — Jitter as Structural Information (Level I)	9
10.1	A1 — Jitter Discrimination Threshold	10
10.1.1	Hypothesis	10
10.1.2	Falsifiable prediction	10
10.1.3	Null hypothesis	10
10.1.4	Design	10
10.1.5	Analysis	11
10.2	A2 — Marine Coherence Predicts Auditory Salience	11
10.2.1	Hypothesis	11
10.2.2	Falsifiable prediction	11
10.2.3	Null hypothesis	11
10.2.4	Design	11
10.2.5	Analysis	12
10.3	A2b — Harmonic-Context Replication (Secondary Confirmatory)	12
10.3.1	Rationale	12
10.3.2	Falsifiable prediction (identical criterion to A2)	12

10.3.3	Null hypothesis	12
10.3.4	Design	12
11	Prediction Set B — HCF1 Compression Architecture (Level II)	13
11.1	B1 — Harmonic Envelope Conformity	13
11.1.1	Hypothesis	13
11.1.2	Falsifiable prediction	13
11.1.3	Null hypothesis	13
11.1.4	Design	13
11.1.5	Analysis	14
11.2	B2 — HCF1 Bit Reduction and Perceptual Transparency	14
11.2.1	Hypothesis	14
11.2.2	Falsifiable prediction	14
11.2.3	Null hypotheses	14
11.2.4	Design	14
11.2.5	Analysis	15
12	Prediction Set C — MEM 8 Associative Memory Architecture (Level III)	15
12.1	C1 — Quadratic Storage Capacity via Phase Encoding	15
12.1.1	Hypothesis	15
12.1.2	Falsifiable predictions	15
12.1.3	Null hypothesis	15
12.1.4	Design	15
12.1.5	Analysis	16
12.2	C2 — Resistance to Catastrophic Forgetting	16
12.2.1	Hypothesis	16
12.2.2	Falsifiable predictions	16
12.2.3	Null hypothesis	17
12.2.4	Design	17
12.2.5	Analysis	17
13	Interpretation of Negative Results	17
14	Replication Pathways	19
14.1	Accessibility by prediction	19
14.2	Recommended entry points for independent labs	19
14.3	Recommended datasets	19
15	Summary of Pre-Registered Predictions	20
16	Standard Pre-Registration Boilerplate	20
16.1	IRB and ethics	21
16.2	Data availability statement	21
16.3	Deviations from the pre-registered plan	21
16.4	Conflicts of interest	21
16.5	Suggested repositories	21

Notation and Terminology

The following definitions are fixed for the entirety of this document. Terminological consistency matters for pre-registrations because post-hoc reinterpretation of key terms can obscure whether a prediction was confirmed or disconfirmed.

Temporal jitter. Cycle-to-cycle variation in the period of a quasi-periodic waveform, quantified as $J = \sigma(T) / \mu(T)$, where T denotes the set of measured inter-peak intervals over a specified time window (default: 200 ms). Jitter is expressed as a percentage of mean period.

Marine coherence metric (κ). A normalised inverse-jitter quantity defined as $\kappa = 1 / (J + \varepsilon)$, where $\varepsilon > 0$ is a small regularising constant (default: 10^{-3}). Higher κ indicates more temporally stable oscillation. The “Marine” qualifier refers to the salience algorithm in which this metric originates; it does not carry implications about consciousness or authenticity.

Spectral prominence (P). Level of a partial above the local noise floor, measured in dB: $P = L_{\text{partial}} - L_{\text{noise}}$.

HCF1 (Harmonic Conformity Flag, 1-bit). A compression scheme storing a master amplitude envelope for the fundamental (e_1) and, for each upper harmonic $h = 2, \dots, H$, a single bit indicating conformity to a pre-calibrated linear prediction $\hat{e}_h = a_h e_1 + b_h$. Non-conforming frames store an explicit residual.

MEM|8 (Modulated Encoding Memory, 8 wave functions per node). A proposed associative memory architecture in which pairwise associations are encoded as phase relationships among complex sinusoidal wave functions. The hypothesis is that N wave functions can store $O(N^2)$ associations, contrasting with the $O(N)$ capacity of standard Hopfield networks.

Constraint horizon. The requirement that any empirical claim specify in advance what evidence would count against it. A framework that can accommodate every possible outcome is not a scientific theory; it is a descriptive vocabulary. All six predictions in this package are designed to have a clear constraint horizon.

Retention (%). In sequential learning experiments (C2): the ratio of recall accuracy after learning a second association set to recall accuracy before, expressed as a percentage.

Confirmatory vs. exploratory. Analyses declared in this document before data collection are *confirmatory*. Analyses conducted after data collection, whether planned or opportunistic, are *exploratory* and must be labelled as such in any resulting publication.

Throughout this document, notation is standardised as follows:

Symbol / term	Meaning
f_0	Fundamental frequency
h	Harmonic index ($h = 1 = \text{fundamental}$)
$e_h(t)$	Amplitude envelope of harmonic h at time t
J	Jitter (normalised period SD)
κ	Marine coherence metric
P	Spectral prominence (dB)
d'	Sensitivity index (signal detection theory)
R^2	Coefficient of determination
ΔR^2	Incremental explained variance
N	Number of wave functions in MEM 8
M	Number of items forming M^2 pairwise associations
MEM 8	Modulated Encoding Memory architecture
HCF1	Harmonic Conformity Flag (1-bit) compression scheme

Scientific Positioning

The predictions in this package sit within three established research traditions. Anchoring them explicitly prevents the framework’s novel terminology from appearing to arise in a vacuum.

Prediction Set A — Psychoacoustics and Temporal Coding

The question of whether fine temporal structure contributes to auditory perception beyond what is captured by the spectral envelope has a long history. Phase locking in the auditory nerve supports temporal coding up to approximately 4–5 kHz (Rose et al., 1967; Palmer & Russell, 1986). The role of micro-temporal variation in the perception of naturalness and timbral identity has been studied in the context of vibrato, mechanical fluctuations in musical instruments, and the perceptual consequences of phase randomisation (McAdams, 1984; Beauchamp, 1993). Auditory salience models typically operationalise salience in terms of spectral energy, spectral flux, or onset strength (Kayser et al., 2005; Klapuri, 2006); the Marine coherence metric represents a proposal that temporal stability constitutes an orthogonal salience dimension not captured by these spectral proxies. Prediction A2 tests this directly as an incremental regression effect.

Prediction Set B — Perceptual Audio Coding and Harmonic Modelling

Harmonic signal models underlie a broad family of audio codecs and analysis methods, including sinusoidal modelling (McAulay & Quatieri, 1986), source-filter approaches to speech coding (Rabiner & Schafer, 1978), and structured prediction in transform codecs. The observation that harmonic amplitude envelopes co-vary across a note (Prediction B1) is related to but distinct from standard spectral envelope coding: HCF1 exploits inter-harmonic temporal correlation rather than across-frequency spectral shape. In this sense it is closer in spirit to predictive coding of envelope trajectories (Spanias, 1994) than to masked-band discard strategies used in perceptual codecs such as MP3 and AAC.

Prediction Set C — Associative Memory and Attractor Networks

Hopfield networks (Hopfield, 1982) are the canonical model of content-addressable memory with a known storage capacity of approximately $0.138N$ patterns for N binary units (Amit et al., 1985). Complex-valued and phase-coded extensions of Hopfield-type networks have been studied theoretically (Noest, 1988; Jankowski et al., 1996; Aoyagi, 1995) and motivate the claim that phase relationships among oscillatory units can encode pairwise associations more efficiently than binary state vectors. The MEM|8 architecture is best understood as a proposal within this family of complex-valued associative networks. Predictions C1 and C2 are designed to be interpretable within this existing literature: the quadratic capacity claim (C1) is a strong quantitative prediction relative to known Hopfield scaling, and the catastrophic forgetting result (C2) connects to the well-studied stability-plasticity dilemma (Grossberg, 1987; McCloskey & Cohen, 1989).

Cross-Level Architectural Unity

Although Prediction Sets A, B, and C address distinct experimental domains, they are unified by a single underlying principle: *structured redundancy under physical constraint*.

Level I (Prediction Set A) concerns the **extraction** of structurally informative temporal patterns from harmonic signals. The claim is that jitter is not noise to be discarded but signal to be retained because it encodes the physical coupling state of the oscillating system.

Level II (Prediction Set B) concerns the **representation** of correlated harmonic structure. Given that upper harmonics are driven by shared physical dynamics, their amplitude envelopes are expected to be highly predictable from the fundamental. HCF1 exploits this predictability to achieve compact, lossless (within-frame) description of harmonic temporal evolution.

Level III (Prediction Set C) concerns **retrieval** under structured relational encoding. If associative memory is implemented via phase relationships among oscillatory units, then the

information-theoretic capacity of the representation scales differently from binary state vectors — specifically, quadratically with the number of wave functions.

All three levels therefore concern constrained representation under physically or structurally motivated redundancy. The framework’s contribution, if confirmed, would be identifying that this principle applies coherently across the domains of perception (Level I), compression (Level II), and memory (Level III).

It is important to note that the three levels are architecturally related but *empirically independent*. Falsification of one prediction set does not imply falsification of the others. See Section 13 for a systematic treatment of what each negative result would and would not entail.

Explainability as an Independent Contribution

A common implicit assumption in systems like MEM|8 is that the primary measure of success is capacity or efficiency relative to existing architectures. This framing sets a high bar: if the quadratic capacity claim in C1 is not confirmed, the architecture might appear to offer nothing. This section records a secondary contribution claim that does *not* depend on the capacity results.

Phase-based associative memory encodes relationships as geometric objects in the complex plane. Each stored association (i, j) corresponds to an angular offset $\Delta\phi_{ij}$ between two wave functions. This has two interpretive consequences that standard binary Hopfield networks do not share:

1. **Geometric interpretability.** The strength of an association can be read directly from the phase offset magnitude; the direction of retrieval (from i to j vs. j to i) corresponds to the sign of the angle. A trained weight matrix in a binary Hopfield network offers no analogous geometric decomposition without additional analysis.
2. **Graded similarity.** Items with similar phase profiles are geometrically close in the complex plane, enabling a natural similarity metric without additional distance computation. This may be useful for approximate nearest-neighbour retrieval in applications where exact matches are rare.

These properties hold regardless of whether the capacity results in C1 are confirmed. If C1 fails — if the quadratic scaling does not materialize — the architecture still offers a transparent, geometrically interpretable representation of associative structure that may be preferable to binary attractor networks in settings where interpretability matters (e.g., embedded systems, explainable AI pipelines, or educational demonstrations of associative memory).

This claim is explicitly *not* pre-registered as a falsifiable prediction in the present document, because interpretability is a design property, not an experimental outcome. It is recorded here so that the broader value of the MEM|8 architecture is not contingent on the specific capacity thresholds in C1.

Operational Boundary Conditions

A well-bounded framework is more tractable, not weaker. The following are explicit non-claims of the Salience-Centered Paradigm as operationalized in this package.

- B1. No claim to proof of consciousness.** Nothing in Levels I–III entails or supports claims about conscious experience, phenomenal authenticity, or subjective perception of “cognitive field” effects. The Marine coherence metric is a signal-processing quantity.
- B2. No rejection of standard psychoacoustic masking.** The framework does not claim that psychoacoustic masking models (as used in MP3, AAC, and related codecs) are incorrect. Prediction A1 is compatible with masking-based models; it tests whether jitter carries additional perceptual information *beyond* masked-band predictions.
- B3. No claim of current engineering superiority.** HCF1 compression is not claimed to out-

perform existing perceptual codecs in general. Prediction B2 tests a specific, narrow claim about bitrate reduction for harmonic envelope coding, not end-to-end audio quality.

B4. No claim of biological plausibility for MEM|8. The MEM|8 architecture is a mathematical model. Predictions C1 and C2 are claims about a defined computational system, not about neural implementation in biological tissue.

B5. No claim that Level IV is false. This document takes no position on whether Level IV claims about consciousness and authenticity are correct. They are excluded solely because they lack sufficient operationalization to support pre-registered experimental design at this time.

B6. No claim that confirmation of Levels I–III entails Level IV. Even if all six predictions are confirmed, that result would not constitute evidence for the Level IV speculative claims. The inferential boundary is sharp.

Statistical Philosophy and Confirmatory Logic

Confirmatory versus exploratory analyses

This package follows the *strong* form of pre-registration: all confirmatory analyses are fully specified prior to any data collection, including dependent variables, null hypotheses, statistical tests, alpha levels, sample sizes, and exclusion criteria. Analyses not specified here are *exploratory* by default and must be labelled as such. Exploratory analyses are scientifically valuable and are not discouraged; the distinction matters solely for inferential status.

Effect thresholds and estimation

Null hypothesis tests are supplemented by explicit effect thresholds wherever possible (e.g., $d' > 0.5$, $\Delta R^2 \geq 0.15$, retention $\geq 90\%$). These thresholds are motivated by theoretical and practical significance, not by statistical convenience. The goal is to ask not only whether an effect exists but whether it is large enough to matter for the claims of the paradigm.

Signal detection theory (Predictions A1, A2)

Signal detection theory (SDT) provides the appropriate framework for perceptual discrimination studies because it dissociates sensitivity (d') from response bias (c). A listener who is more willing to report jitter (liberal criterion) will produce more hits but also more false alarms; d' is unaffected by this bias. This is important for Prediction A1 because motivation, attention, and individual criterion placement could otherwise confound a simple proportion-correct measure.

Incremental regression (Prediction A2)

The decision to test ΔR^2 rather than a raw correlation reflects the incremental logic of the Marine coherence hypothesis: the claim is that coherence predicts salience *beyond* spectral prominence, not that it predicts salience in isolation. A model comparison that does not control for prominence would not test the paradigm's actual claim.

Replication priority

Narrative coherence of the paradigm is explicitly subordinated to replication priority. If a prediction fails, the result stands. Post-hoc reinterpretation of failed predictions as consistent with the paradigm is not permitted within the confirmatory framework; it must be declared exploratory.

Relation to Existing Compression Paradigms

The HCF1 scheme (Prediction Set B) is most clearly understood by contrast with existing approaches to audio compression.

Paradigm	What is discarded / predicted	Contrast with HCF1
Transform coding (MP3, AAC)	Masked spectral bands; fine temporal structure	HCF1 retains all temporal structure; no masking model used
Linear predictive coding (LPC)	Spectral envelope; residual coded separately	HCF1 operates on amplitude envelopes, not the waveform directly
Sinusoidal modelling	Non-sinusoidal residual; parameters tracked per partial	HCF1 exploits inter-harmonic envelope correlation, not per-partial tracking
Predictive envelope coding	Temporal redundancy across frames	HCF1 exploits redundancy <i>across harmonics</i> within a frame

Table 1: Conceptual comparison of HCF1 with established compression paradigms. The distinguishing feature of HCF1 is the hypothesis that upper harmonic amplitude envelopes are predictable from the fundamental with sufficient accuracy to permit 1-bit flag encoding. This is an empirical claim (Prediction B1) that must be confirmed before the compression scheme can be evaluated (Prediction B2). The two predictions are therefore logically ordered: B1 is the necessary prior condition for B2.

Computational Complexity and Scalability of MEM|8

Prediction Set C involves a computational architecture that may be unfamiliar to readers outside the associative memory literature. The following summary clarifies the relevant asymptotic properties and their experimental implications.

Storage complexity

A MEM|8 system with N wave functions has two distinct storage components. First, the N oscillator states $\{\phi_k\} \in \mathbb{C}^N$ contribute $O(N)$ degrees of freedom. Second, an antisymmetric pairwise offset matrix $\Delta_{ij} \in \mathbb{R}^{N \times N}$ (zero diagonal) contributes $O(N^2)$ entries — one target angular offset per ordered pair. It is this matrix, not the oscillators alone, that provides the hypothesised relational capacity. The claim that $O(N^2)$ associations can be stored is therefore a claim about constraint-satisfaction in the joint oscillator–offset system, not a claim that phase continuity alone implies quadratic capacity.

This distinction matters for interpretation: if C1 is confirmed, the result would suggest that a phase-relational constraint system exploits pairwise angular structure efficiently and exhibits sharp retrieval transitions under interference. If C1 fails, the failure would most likely indicate that the relaxation algorithm cannot simultaneously satisfy $O(N^2)$ constraints given only $O(N)$ oscillator degrees of freedom.

For comparison, the standard Hopfield network stores at most $\approx 0.138N$ patterns reliably before interference dominates (Amit et al., 1985).

Retrieval complexity

Retrieval in MEM|8 involves nearest-neighbour search in phase space. For a full pairwise comparison this is $O(N^2)$, though approximate retrieval schemes could reduce this. Predictions C1 and C2 are agnostic to the retrieval algorithm; they concern accuracy at convergence, not speed.

Robustness under noise

A known limitation of phase-coded associative memories is sensitivity to phase noise: small errors in the representation of wave functions can produce retrieval failures. The simulations in Predictions C1 and C2 use floating-point arithmetic and therefore understate real-world noise sensitivity. Any practical deployment would require additional robustness analysis. The predictions in this package are therefore best understood as establishing *upper bounds* on capacity and retention under idealised conditions.

Scalability

Experiments C1 and C2 use $N \in \{64, 128, 256\}$. These are toy-scale models. Whether the claimed advantages persist at $N \gtrsim 10^4$ (required for practical application) is an open question that lies beyond the scope of this pre-registration.

Preamble and Epistemic Framework

The predictions collected in this package are derived from the technically grounded strata of the Saliency-Centered Paradigm. The organizing principle throughout is the *constraint horizon*: a well-formed empirical claim must specify in advance what evidence would count against it. The framework generates these predictions not as confirmatory demonstrations but as genuine empirical risks. A prediction set whose failure would be shrugged off is not a scientific prediction; each of the six templates below is designed so that a negative outcome would carry clear evidential weight against the corresponding level of the paradigm.

Level	Core claim	Status
I	Temporal jitter encodes structural information beyond masking	Testable; Prediction Set A
II	Harmonic envelopes are linearly predictable from the fundamental; 1-bit flag compression is efficient	Testable; Prediction Set B
III	Phase-based associative memory yields quadratic capacity and resists catastrophic forgetting	Testable; Prediction Set C
IV	Ultrasonic frequencies alter brain coherence; spectral fracture produces “authenticity loss”	<i>Excluded — requires prior operationalization</i>

Data availability. All stimuli, code, and anonymized data will be deposited to an open repository (OSF, Zenodo, or equivalent) upon study completion. Deviations from the pre-registered analysis plan must be declared in print.

Conflicts of interest. The author has a theoretical investment in the Saliency-Centered Paradigm. Independent replication is therefore especially encouraged.

Prediction Set A — Jitter as Structural Information (Level I)

A1 — Jitter Discrimination Threshold

Template A1

Full title. Detection of temporal jitter in harmonic complexes matched for spectral envelope and pitch.

Hypothesis

Listeners can discriminate natural jitter (period variation $\approx 0.5\text{--}3\%$) from perfectly periodic harmonics when spectral envelope, fundamental frequency, and amplitude envelope are held constant. This tests whether jitter constitutes an audible signal dimension orthogonal to spectrum and amplitude.

Falsifiable prediction

Primary (confirmatory): Mean $d' > 1.0$ across participants (one-tailed t -test against 1.0; $\alpha = 0.05$, power = 0.80). A threshold of $d' = 1.0$ corresponds to approximately 76% correct in 2AFC — a level of sensitivity that requires jitter to be more than marginally detectable. The lower value of $d' > 0.5$ ($\approx 60\%$ correct) was rejected during design because it is consistent with chance-level performance in many auditory discrimination studies and would not demonstrate that jitter carries practically meaningful signal.

Secondary (confirmatory): d' increases monotonically with jitter magnitude across three pre-specified levels (0.5%, 1.5%, 2.5% RMS period variation). This is tested via a one-way repeated-measures ANOVA with a significant linear contrast ($p < 0.05$). Both criteria must be met for the prediction to be considered confirmed.

Null hypothesis

Mean $d' \leq 1.0$, or no monotonic relationship between d' and jitter magnitude.

Design

Stimuli. Synthesized harmonic complex ($f_0 = 220$ Hz, harmonics 1–10, equal amplitude modulo a fixed spectral envelope, duration 500 ms, $\cos^2$ onset/offset ramps of 10 ms).

- **Condition A (three levels).** Temporal jitter is generated algorithmically (see below) at RMS period variation of 0.5%, 1.5%, and 2.5% of mean period. Applied by resynthesis from a period sequence; no recorded audio is used directly.
- **Condition B.** Identical resynthesis procedure with all period deviations set to zero (perfectly isochronous).

Jitter synthesis procedure. For target RMS jitter $J \in \{0.005, 0.015, 0.025\}$: generate $K = 111$ period values $T_k = T_0(1 + \epsilon_k)$, where $T_0 = 1/220$ s, $\epsilon_k \sim \mathcal{N}(0, J^2)$, clipped to $[-3J, 3J]$. Apply a 2nd-order Butterworth low-pass filter at 30 Hz to ϵ_k before use (removes inaudibly fast variation while preserving temporal fluctuation in the audible jitter range). Resynthesize each period as one cycle of the harmonic complex using randomized starting phases (per-file seed, fixed across conditions). Phase randomization controls waveform morphology and fine temporal microstructure across conditions but does not equate spectral envelopes, which are explicitly matched. Stimuli are generated deterministically from a fixed seed list; synthesis code is deposited to OSF prior to data collection.

Period estimation (for replicator verification): Jitter magnitude in synthesized stimuli should be verified using an autocorrelation-based or Hilbert instantaneous-phase period tracker, not raw peak detection. Peak detection in complex harmonic stimuli is unreliable because harmonic interference displaces apparent peak locations. The verification criterion is: measured J within ± 0.002 of target J .

Task. Two-interval, two-alternative forced choice (2I-2AFC) with corrective feedback after each trial. Each interval is 500 ms, inter-stimulus gap 300 ms. X drawn uniformly from $\{A, B\}$; participant indicates which interval contained jitter.

Participants. $N = 30$ normal-hearing adults (audiometric threshold ≤ 20 dBHL at

0.5–4 kHz). Power analysis: $\Delta d' = 1.0$ within-subjects design, 90% power implies $n \approx 22$; 35% buffer applied.

Trials. 120 per participant (30 per jitter level $\times$ 3 levels + 30 jitter-absent), randomized without replacement per block of 12.

Blinding. The experimenter running sessions is blind to the jitter magnitude assigned to each stimulus file. Stimulus files are labelled by a numeric code assigned by a separate researcher. Analysis is conducted on anonymized data files before the code is broken.

Analysis

Sensitivity is computed as:

$$d' = z(H) - z(F)$$

where H is the hit rate and F is the false alarm rate. Signal detection theory is used because d' is invariant to response bias, preventing criterion differences between participants from inflating or suppressing the sensitivity estimate.

Primary: One-sample t -test of mean d' (collapsed across jitter levels) against 1.0, one-tailed.

Secondary: One-way repeated-measures ANOVA on d' at 0.5%, 1.5%, and 2.5% RMS jitter; pre-specified linear contrast.

Exclusion criteria: fewer than 80% of trials completed; accuracy $< 55\%$ on the 10-trial practice block (accuracy is used here rather than d' because the practice block is too short for stable SDT estimation).

Confirmatory: both primary and secondary criteria must be met. **Exploratory:** interaction between jitter magnitude and musical training.

A2 — Marine Coherence Predicts Auditory Salience

Template A2

Full title. Temporal coherence (Marine coherence metric κ) adds incremental predictive power over spectral prominence in auditory salience ratings.

Hypothesis

The Marine coherence metric κ predicts listener salience ratings beyond what is accounted for by spectral prominence alone.

Falsifiable prediction

In a multiple regression, $\hat{\beta}_2 > 0$ with $p < 0.001$, and $\Delta R^2 \geq 0.08$ after adding κ to a model containing only prominence P . The threshold of $\Delta R^2 = 0.08$ reflects published benchmarks in auditory salience and psychoacoustic modelling (e.g., Klapuri, 2006; Kayser et al., 2005) where a single novel predictor accounting for 8–10% of unique variance is considered practically meaningful. A higher threshold ($\Delta R^2 \geq 0.15$) was considered but rejected as poorly calibrated to the expected magnitude of a single jitter-derived metric relative to the strong baseline provided by spectral prominence.

Null hypothesis

$\hat{\beta}_2 \leq 0$ or $\Delta R^2 < 0.04$ (coherence adds no practically meaningful predictive power beyond spectral prominence).

Design

Stimuli. 100 partials extracted from 10 recorded instrumental notes (piano, violin, guitar, voice; notes selected to span the full range of jitter magnitudes observed in pilot measurements). For each partial:

- Prominence P = dB above local noise floor (estimated via median filter of width 100 Hz in log-frequency).
- Marine coherence metric $\kappa = 1/(J + 0.001)$, measured over a 200 ms central window.

Rating task. 20 normal-hearing listeners rate each partial in isolation on a 0–100 continuous slider labelled “barely audible” to “very salient.” Each partial is presented alone (without harmonic context) once, in individually randomized order. This presentation mode is acknowledged as a limitation: partial salience in isolation may differ from salience in a harmonic context. A naturalistic follow-up study is designated exploratory. Mean rating across listeners is the dependent variable.

Blinding. Listeners are blind to all hypotheses. The analyst computing κ is separate from the analyst running the regression, to prevent confirmation bias in feature extraction choices.

Regression models.

$$\text{Model 1: } \hat{S} = \beta_0 + \beta_1 P \quad (1)$$

$$\text{Model 2: } \hat{S} = \beta_0 + \beta_1 P + \beta_2 \kappa \quad (2)$$

The incremental test evaluates whether κ explains variance in salience over and above spectral prominence. This is the appropriate test because the paradigm claims that temporal coherence is an *orthogonal* salience dimension, not a substitute for spectral energy.

Analysis

Linear mixed model with listener as a random intercept, or ordinary least squares on listener-averaged ratings. Report ΔR^2 and F -test for model comparison. Outlier removal: Cook’s $D > 4/n$ for any partial, declared before unblinding.

Confirmatory: $\hat{\beta}_2 > 0$, $p < 0.001$, and $\Delta R^2 \geq 0.08$. **Exploratory:** interaction $P \times \kappa$; instrument-family moderator; naturalistic harmonic-context replication.

A2b — Harmonic-Context Replication (Secondary Confirmatory)

Template A2b

Full title. Marine coherence metric predicts salience of harmonic complexes presented in full harmonic context.

Rationale

A2 presents partials in isolation, which is acknowledged as ecologically limited: temporal coherence effects in auditory perception often require harmonic context to manifest via common-fate grouping and fusion. A2b is a pre-registered secondary confirmatory study using naturalistic stimuli. Both A2 and A2b must be reported; they are independent tests of the same hypothesis under different stimulus conditions.

Falsifiable prediction (identical criterion to A2)

$\hat{\beta}_2 > 0$, $p < 0.001$, and $\Delta R^2 \geq 0.08$.

Additional criterion (demotes to exploratory if violated): The ΔR^2 in A2b must be numerically $\geq$ the ΔR^2 in A2. If context adds no incremental effect, the harmonic-context result is exploratory regardless of significance.

Null hypothesis

$\hat{\beta}_2 \leq 0$, or $\Delta R^2 < 0.08$, or $\Delta R^2_{A2b} < \Delta R^2_{A2}$.

Design

Stimuli. 50 harmonic complexes (harmonics 1–10 simultaneously) from the same 10 instrumental recordings used in A2. Duration 1 s, 10 ms $\cos^2$ ramps.

Feature aggregation. Two pre-registered strategies (both must meet criteria for confirmation):

- **Max rule:** $P_{\text{cmplx}} = \max_h P_h$; $\kappa_{\text{cmplx}} = \max_h \kappa_h$.
- **Energy-weighted mean:** $P_{\text{cmplx}} = \sum_h w_h P_h$; $\kappa_{\text{cmplx}} = \sum_h w_h \kappa_h$, where $w_h = P_h / \sum_h P_h$.

Rating task. 20 normal-hearing listeners (may overlap with A2 sample). 0–100 slider; mean across listeners is the dependent variable.

Analysis. Same mixed model as A2; report ΔR^2 per aggregation rule. Confirmation requires both aggregation rules to meet all three criteria.

Confirmatory: as stated. **Exploratory:** comparison of max vs. weighted-mean ΔR^2 ; partial vs. complex moderation.

Prediction Set B — HCF1 Compression Architecture (Level II)

B1 — Harmonic Envelope Conformity

Template B1

Full title. Amplitude envelopes of upper harmonics are linearly predictable from the fundamental in acoustic instruments.

Hypothesis

For physically coupled harmonic oscillators, the amplitude envelope of each upper harmonic is well approximated by a linear function of the fundamental's envelope. This is the necessary prior condition for the HCF1 compression scheme.

Falsifiable prediction

Median $R^2 > 0.90$ across harmonics $h = 2, \dots, 10$, with the lower bound of the 95% bootstrap confidence interval exceeding 0.80.

Null hypothesis

Median $R^2 < 0.70$ (harmonic-specific variation dominates; envelopes are not jointly predictable from the fundamental).

Design

Recordings. 20 notes per instrument family (different pitches, sampled at 44.1 kHz, minimum duration 2 s): plucked string (classical guitar), struck string (piano), bowed string (violin), struck membrane (timpani).

Analysis procedure.

1. Detect f_0 per frame (10 ms hop, 50% overlap) via an autocorrelation method.
2. Extract amplitude envelope $e_h(t)$ for harmonics $h = 1, \dots, 10$ via an 8th-order Butterworth bandpass filter (bandwidth $\pm 20\%$ of hf_0), followed by full-wave rectification and low-pass filtering at 30 Hz.
3. For each harmonic $h = 2, \dots, 10$, fit by ordinary least squares:

$$\hat{e}_h(t) = a_h \cdot e_1(t) + b_h$$

Compute $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$.

Exclusion: Notes with inharmonicity $> 1\%$ are excluded from the confirmatory sample and reported in a separate table. Inharmonicity is operationally defined as the mean absolute deviation of measured partial frequencies from their predicted values $h \cdot f_0$, normalized by f_0 :

$$\text{Inh} = \frac{1}{H} \sum_{h=1}^H \frac{|\hat{f}_h - h \cdot f_0|}{f_0}$$

where $\hat{f}_h$ is the measured frequency of the h -th partial (estimated via spectral peak picking on a zero-padded FFT). This definition is fixed prior to data collection; the threshold of 1% is pre-registered and cannot be adjusted post-hoc.

A **sensitivity analysis** is mandatory: the confirmatory R^2 test is re-run including all excluded notes. If the result changes sign or crosses the threshold boundary, this must be reported prominently.

Analysis

Median R^2 and 95% bootstrap confidence interval (10 000 resamples over individual recordings). Secondary: distribution of slope a_h — expected to decrease monotonically with h for most instrument families.

Adaptive filter bandwidth (pre-registered exception): If a partial's instantaneous f_0 deviates $> 15\%$ from $h \cdot f_0$ for $> 10\%$ of frames (e.g., notes with wide vibrato or pitch drift), the Butterworth filter bandwidth is widened to $\pm 30\%$ of hf_0 for that partial. The proportion of frames triggering this widening must be reported per note. This exception is declared before analysis; post-hoc bandwidth adjustment is not permitted.

Confirmatory: R^2 threshold. **Exploratory:** differences across instrument families; interaction with note duration and pitch register.

B2 — HCF1 Bit Reduction and Perceptual Transparency

Template B2

Full title. HCF1 with 1-bit conformity flags achieves $\geq 60\%$ bit reduction without audible degradation in a blind listening test.

Hypothesis

Storing one master envelope and a 1-bit conformity flag per harmonic yields substantial bitrate savings while remaining perceptually indistinguishable from the original.

Falsifiable prediction

(1) Bit reduction $\geq 60\%$, **and** (2) ABX discrimination accuracy does not exceed chance: binomial test $p > 0.05$; equivalently, proportion correct < 0.575 (upper 95% one-sided CI for $n = 50, p_0 = 0.5$).

Null hypotheses

(1) Bit reduction $< 40\%$, **or** (2) ABX accuracy $> 57.5\%$.

Design

Compression conditions.

Baseline. Store envelopes $e_1, \dots, e_{10}$ independently at 16 bits, 100 Hz update rate, yielding:

$$R_{\text{baseline}} = 10 \times 16 \times 100 = 16,000 \text{ bits/s}$$

HCF1. Store e_1 at 16 bits, 100 Hz. For $h = 2, \dots, 10$, store one bit per frame:

$$\text{flag}_{h,t} = \begin{cases} 0 & (\text{conform}) \Rightarrow \hat{e}_h(t) = a_h e_1(t) + b_h \\ 1 & (\text{nonconform}) \Rightarrow \text{store residual at 16 bits} \end{cases}$$

Effective bitrate:

$$R_{\text{HCF1}} = \underbrace{16 \times 100}_{\text{fundamental}} + \underbrace{9 \times 1 \times 100}_{\text{9 flag bits/s}} + f_{\text{nc}} \times 9 \times \underbrace{16 \times 100}_{\text{residual/harmonic}} \text{ bits/s}$$

where $f_{\text{nc}} \in [0, 1]$ is the empirically measured nonconformity fraction. At zero nonconformity this reduces to $1600 + 900 = 2500$ bits/s; at full nonconformity it approaches $1600 + 900 + 14400 = 16900$ bits/s ($\approx$ baseline). The 60% reduction criterion requires that the empirically measured f_{nc} yield $R_{\text{HCF1}} \leq 0.4 \times 16000 = 6400$ bits/s.

Listening test. 10 audio files (5–15 s each): solo speech, solo piano, jazz trio, string quartet, pop vocal, acoustic guitar, orchestral excerpt, electronic music, percussion ensemble, mixed ensemble. 20 normal-hearing listeners; 50 ABX trials each. A = original, B = HCF1-reconstructed, X = random draw.

Analysis

Bit reduction computed directly from measured bitrates. ABX: binomial test against $p_0 = 0.5$; non-inferiority margin = 0.075. Pass criterion: reduction $\geq 60\%$ **and** upper CI of proportion correct $< 57.5\%$.

Confirmatory: both criteria jointly. **Exploratory:** effect of musical genre on reduction ratio; correlation between R^2 from B1 and HCF1 efficiency per recording.

Prediction Set C — MEM|8 Associative Memory Architecture (Level III)**C1 — Quadratic Storage Capacity via Phase Encoding****Template C1**

Full title. Sharp retrieval transition in MEM|8: testing predicted dense relational capacity in a phase-based associative memory architecture.

Hypothesis

A MEM|8 system with N wave functions and a pairwise offset matrix Δ_{ij} is hypothesised to support dense relational encoding such that retrieval accuracy remains high below approximately $M^2/N \approx 1$ and degrades sharply near $M^2/N \approx 1$ –2. This is motivated by analogy to constraint satisfaction phase transitions (e.g., random k -SAT), where underconstrained systems admit solutions near-perfectly and overconstrained systems fail sharply. The hypothesis is not that a formal proof of $O(N^2)$ capacity exists, but that the empirical accuracy curve behaves consistently with this scaling regime in simulations up to $N = 256$.

Falsifiable predictions

- Retrieval accuracy $> 80\%$ when $M^2 \leq N$ (underconstrained regime).
- Retrieval accuracy $< 50\%$ when $M^2 > 2N$ (overconstrained regime).
- **Sharpness (pre-registered quantitative criterion):** A logistic function $\hat{a}(x) = 1/(1 + e^{-k(x-x_0)})$ is fit to accuracy as a function of $x = M^2/N$. The transition is considered sharp if the fitted slope parameter $k > 8$ (relaxed from 10 to account for finite-size effects at $N = 64$). Alternatively, if the logistic fit converges poorly ($R_{\text{fit}}^2 < 0.80$), the bootstrapped accuracy difference $a(M^2 = N) - a(M^2 = 2N) > 0.4$ is used. The primary criterion (logistic slope) takes precedence; the fallback is declared before analysis.

These thresholds are empirical predictions, not mathematical guarantees. Random constraint systems can become frustrated before naive counting thresholds suggest, and finite-size effects may shift the observed transition. The predictions specify what the simulations *should* show if the architecture behaves as hypothesised; deviations are informative regardless of direction.

Null hypothesis

Accuracy scales linearly with N (no quadratic advantage), or $k \leq 5$ in the logistic fit (gradual degradation rather than sharp phase transition).

Design

Implementation. Floating-point (double precision) simulation of MEM|8 with $N \in \{64, 128, 256\}$. The architecture uses two components: (1) N wave functions $\psi_k = e^{i\phi_k}$ on the unit circle, and (2) an antisymmetric offset matrix $\Delta \in \mathbb{R}^{N \times N}$ (zero diagonal) storing pairwise target offsets. It is the offset matrix, not the oscillator states alone, that encodes associations.

MEM|8 pseudocode (corrected, replication-ready):*Initialization:*

- $\phi_k \sim \text{Uniform}[0, 2\pi)$ for $k = 1, \dots, N$.

- $\Delta_{ij} \leftarrow 0$ for all i, j ; $\Delta_{ii} = 0$ (enforced).

Storage of association (i, j) :

1. Record target offset: $\Delta_{ij} \leftarrow \angle(\psi_j / \psi_i)$; $\Delta_{ji} \leftarrow -\Delta_{ij}$ (antisymmetric).
2. Run relaxation (10 iterations, fixed):
 - a. Compute current pairwise angles: $C_{ij} = \angle(\psi_j / \psi_i)$ for all (i, j) .
 - b. Error matrix: $E_{ij} = \angle(e^{i(\Delta_{ij} - C_{ij})})$ (wrapped to $[-\pi, \pi]$).
 - c. Update each wave function: $\phi_k \leftarrow \eta \cdot \text{clip}\left(\frac{1}{N} \sum_j E_{kj} + \frac{1}{N} \sum_i E_{ik}, -0.5, 0.5\right)$ with learning rate $\eta = 0.1$.
 - d. Normalize: $\psi_k \leftarrow \phi_k / |\phi_k|$.

Retrieval given cue i :

1. For each candidate $j \neq i$, compute predicted phase: $\hat{\phi}_j = \phi_i + \Delta_{ij}$.
2. Angular distance: $d_j = |\angle(e^{i(\hat{\phi}_j - \phi_i)})|$.
3. Retrieved item: $j^* = \arg \min_{j \neq i} d_j$.
4. Correct if $j^* = j_{\text{true}}$.

This specification is complete and self-contained. Any implementation that deviates from this pseudocode — including different relaxation iterations, learning rate schedules, or distance metrics — constitutes a variant and must be so described.

Load conditions. $M \in \{\lfloor \sqrt{N/2} \rfloor, \lfloor \sqrt{N} \rfloor, \lfloor \sqrt{2N} \rfloor, \lfloor \sqrt{4N} \rfloor\}$, yielding four load regimes per N . Association pairs are drawn uniformly at random from the N item slots without replacement within each set.

Trials. 100 independently generated random association sets per (N, M) condition.

Analysis

Accuracy plotted as a function of M^2/N with 95% bootstrap CI over the 100 association sets. Logistic regression fit to the accuracy curve; slope k extracted and tested against the pre-registered threshold of 8.

Validation test (mandatory before full C1 run): Run $N = 16$, $M^2/N \in \{0.25, 0.5, 1.0, 1.5, 2.0, 3.0\}$, 20 seeds each. Plot accuracy vs. M^2/N . If no transition is visible (accuracy remains constant $> 80\%$ or $< 50\%$ throughout), the implementation is incorrect and must be debugged before proceeding.

Confirmatory: accuracy thresholds at $M^2 \approx N$ and $M^2 \approx 2N$; logistic sharpness criterion $k > 8$. **Exploratory:** effect of phase quantisation (fixed-point vs. floating-point); comparison to complex-valued Hopfield networks (Noest, 1988; Jankowski et al., 1996) on the same task.

C2 — Resistance to Catastrophic Forgetting

Template C2

Full title. MEM|8 retains prior associations after sequential learning, unlike Hopfield networks at equivalent load.

Hypothesis

MEM|8 shows $\leq 10\%$ drop in recall accuracy after learning a second association set; a standard Hopfield network at the same load drops $\geq 50\%$. This tests whether the phase-encoding strategy confers a stability-plasticity advantage.

Falsifiable predictions

- MEM|8 retention $\geq 90\%$ of original accuracy.
- Hopfield retention $\leq 50\%$ of original accuracy.

- Difference significant at $p < 0.001$ (two-sample Welch t -test).

Null hypothesis

MEM|8 retention $< 80\%$, or not statistically different from Hopfield at $\alpha = 0.001$.

Design

Networks. MEM|8 ($N = 256$ complex wave functions, retrieval as specified in C1) vs. a **pair-association Hopfield baseline** ($N = 256$ binary units).

The Hopfield baseline is implemented as a pair-association task, not a pattern completion task, to ensure the comparison is fair. Concretely: each item i is represented as a random $N/2$ -bit binary vector $\mathbf{v}_i$; each pair (i, j) is stored as the outer-product Hebbian update on the weight matrix $W += \mathbf{v}_i \mathbf{v}_j^T + \mathbf{v}_j \mathbf{v}_i^T$. Retrieval: given cue $\mathbf{v}_i$, one step of synchronous update yields $\hat{\mathbf{v}}_j = \text{sgn}(W \mathbf{v}_i)$; correct if nearest to $\mathbf{v}_j$. This formulation is equivalent to standard Hopfield pattern completion on the concatenated pair representation but expressed in the association-retrieval frame.

Both architectures are therefore solving the *same task* under the same performance metric (proportion of correctly retrieved partners), differing only in their internal representations.

Procedure.

1. Learn association set A ($M = 16$ pairs); measure accuracy on A .
2. Learn association set B ($M = 16$ new, non-overlapping pairs, using the same update rule without resetting the weight matrix / phase registers).
3. Retest accuracy on A (same cues as step 1).

$$\text{Retention} = \frac{\text{accuracy on } A \text{ after step 3}}{\text{accuracy on } A \text{ after step 1}} \times 100\%$$

Repetitions. 50 independent random seeds per network type (different random item vectors / phase initializations).

Analysis

Two-sample Welch t -test comparing MEM|8 vs. Hopfield retention distributions. Also report: proportion of seeds where MEM|8 exceeds Hopfield by $> 20\%$. Scatter plot of original vs. retained accuracy (one point per seed, per network).

Confirmatory: retention thresholds and network comparison. **Exploratory:** retention as a function of increasing load ($M \in \{32, 64\}$); comparison to other continual learning baselines (e.g., Elastic Weight Consolidation).

Interpretation of Negative Results

A pre-registration that does not specify the consequences of falsification is incomplete. The following table states what each negative outcome would and would not entail.

Pred.	If falsified (negative outcome)	Implications and limits
A1	Listeners cannot detect jitter at $d' > 1.0$, or d' does not increase monotonically with jitter magnitude	Weakens the claim that jitter encodes perceptually salient structural information at practically meaningful levels. Note: $d' > 0.5$ (weak detection) might still occur; the higher threshold is deliberately set to avoid claiming support from marginal sensitivity. Does not falsify B or C.
A2	κ adds no salience variance ($\Delta R^2 < 0.08$, $p \geq 0.001$)	Weakens the Marine coherence metric as a perceptual predictor. Does not falsify the compression claim (B) or memory claim (C). Would suggest κ is an engineering quantity without direct perceptual analogue, but does not preclude its use as a compression or detection feature.
B1	Harmonic envelopes are not linearly predictable ($R^2 < 0.70$)	Directly undermines the conformity assumption behind HCF1. Makes B2 theoretically unmotivated. Does not affect Level I (A) or Level III (C) predictions.
B2	HCF1 achieves $< 40\%$ reduction or is audibly distinguishable	Falsifies the specific engineering claim; does not falsify B1. May indicate that the conformity fraction is instrument- or genre-specific rather than general.
C1	No quadratic capacity advantage; logistic slope $k \leq 5$; or accuracy does not drop below 50% at $M^2 = 2N$	Weakens the central computational claim of MEM 8. Does not affect Level I or Level II predictions. Even if C1 fails, the geometric interpretability of phase-based encoding (Section 4) remains a valid architectural property.
C2	MEM 8 shows comparable forgetting to Hopfield	Weakens the stability-plasticity claim. Combined with C1 failure, would substantially undermine the Level III architecture as specified. Would not affect psychoacoustic or compression predictions.

Table 2: Predicted consequences of falsification for each pre-registered prediction. Predictions are empirically independent; falsification of one does not entail falsification of others.

Independence of levels. The three prediction sets are architecturally related (Section 3) but empirically independent. A complete failure of Set C would leave Sets A and B entirely untouched, and vice versa. This modularity is a deliberate feature: it means the package generates genuine evidence regardless of which predictions succeed or fail.

Replication Pathways

Accessibility by prediction

Pred.	Equipment needed	Software	Est. runtime
A1	Audiometric booth; head-phones; standard PC	Python (NumPy, SciPy, PsychoPy) or MATLAB	$\approx$ 3 h data collection; 30 min analysis
A2	Recordings of acoustic instruments; standard PC	Python (librosa, statsmodels) or R	$\approx$ 1 h data collection; 1 h analysis
B1	Recordings (publicly available datasets acceptable)	Python (librosa, SciPy) or MATLAB	$\approx$ 2 h analysis per instrument family
B2	Audiometric booth; head-phones; standard PC	Python (SciPy, soundfile, PsychoPy)	$\approx$ 3 h data collection; 1 h analysis
C1	Standard PC (no GPU required)	Python (NumPy, SciPy)	$\approx$ 2 h total (all N and M conditions)
C2	Standard PC (no GPU required)	Python (NumPy)	$\approx$ 30 min total

Table 3: Replication requirements by prediction. C1 and C2 are purely computational and require no human participants or specialist equipment.

Recommended entry points for independent labs

Predictions C1 and C2 are the easiest entry points for independent replication: they require only a standard Python environment, no human participants, no IRB approval, and have runtime measured in minutes to hours. Predictions B1 is the next most accessible, as it requires only existing recordings (freely available datasets such as the University of Iowa Musical Instrument Samples library are suitable). Predictions A1, A2, and B2 require a listening environment; a quiet room with calibrated headphones is sufficient for A1 and A2, though an audiometric booth is preferred.

Recommended datasets

- **B1, A2 (instrumental recordings):** University of Iowa Electronic Music Studios Musical Instrument Samples; NSynth (Engel et al., 2017); RWC Music Database (Goto et al., 2003).
- **B2 (audio quality):** EBU SQAM test sequences; internally generated stimuli following the compression procedure specified above.

- **A1 (jitter extraction):** Any high-quality close-microphone recording of an acoustic guitar; jitter parameters specified in Template A1.

Summary of Pre-Registered Predictions

Pred.	Dependent variable	Statistical criterion	Sample	Suggested venue
A1	Sensitivity index d' + monotonic magnitude effect	$d' > 1.0$; linear contrast $p < 0.05$	30 listeners	JASA; Acta Acust.
A2	$\Delta R^2, \hat{\beta}_2$ (isolated partials)	$\Delta R^2 \geq 0.08, p < 0.001$	100 partials, 20 listeners	J. Acoust. Soc.
A2b	$\Delta R^2, \hat{\beta}_2$ (harmonic context)	$\Delta R^2 \geq 0.08$, both aggregation rules	50 complexes, 20 listeners	J. Acoust. Soc.
B1	Median R^2 (harmonic conformity)	$R^2 > 0.90$ (95% CI > 0.80)	80 recordings	ICASSP; EURASIP
B2	Bit reduction + ABX accuracy	$\geq 60\%$ reduction & correct $< 57.5\%$	10 files, 20 listeners	AES; IEEE TASLP
C1	Retrieval accuracy vs. load	$> 80\%$ at $M^2 \leq N$; $< 50\%$ at $M^2 > 2N$; slope $k > 8$	100 sets per (N, M)	Neural Comp.; NeurIPS
C2	Retention (%) after sequential learning	$\text{MEM} 8 \geq 90\%$; Hopfield (corrected) $\leq 50\%$; $p < 0.001$	50 seeds per network	Neural Comp.; ICLR

Table 4: Summary of pre-registered predictions with current statistical criteria, sample sizes, and suggested venues. All thresholds reflect revisions made prior to data collection; the version history is recorded in the OSF repository.

Shared methodological properties

All six predictions satisfy the following requirements for confirmatory pre-registration:

- **Falsifiable.** An explicit null hypothesis is stated for each, identifying what negative evidence would look like.
- **Operationalized.** Key variables are defined with explicit measurement procedures: d' , κ , R^2 , bit reduction, retrieval accuracy, retention.
- **Scope-limited.** No prediction appeals to “consciousness,” “authenticity,” or “cognitive field” — constructs that remain unoperationalized at Level IV.
- **Reproducible.** Each study is executable by an independent laboratory without privileged access to proprietary instruments or data.
- **Modular.** The prediction sets are empirically independent; see Section 13 for a systematic account of what each negative outcome would and would not entail.

Standard Pre-Registration Boilerplate

IRB and ethics

Studies involving human participants (A1, A2, B2) require approval from an Institutional Review Board or equivalent ethics committee before data collection. Participants must provide written informed consent and be compensated at a rate consistent with local norms. Auditory stimuli should be presented at ≤ 70 dB SPL for all listening experiments to avoid risk of hearing damage. Computational studies (B1, C1, C2) do not involve human participants and do not require IRB review.

Data availability statement

Upon completion of each study, the following materials will be deposited to an open repository (OSF, Zenodo, or equivalent under a CC-BY licence):

- All stimuli, or synthesis code sufficient to reproduce them exactly.
- Raw participant data, anonymized in compliance with applicable regulations.
- Analysis scripts (Python, R, or MATLAB, with inline comments).
- This pre-registration document, version-controlled with commit hash.

Deviations from the pre-registered plan

Any deviation from the analysis pipeline specified above must be declared explicitly in the manuscript, with a statement distinguishing confirmatory (pre-registered) from exploratory (post-hoc) findings. Exploratory analyses are encouraged and should be reported transparently; the distinction is not a judgment of scientific value but a statement of inferential status.

Conflicts of interest

The author has a prior theoretical investment in the Salience-Centered Paradigm. Independent replication is therefore especially valuable, and replication teams are encouraged to file their own pre-registration citing this document as the source of the original predictions.

Suggested repositories

Pre-registrations may be filed at:

- [Open Science Framework \(OSF\)](#) — recommended for the full package, as it supports version control and time-stamped registration.
- [AsPredicted](#) — suitable for individual templates (A1–C2) filed separately.
- Journal-specific Registered Reports tracks at *JASA*, *Psychological Science*, *Neural Computation*, or *ICLR*.